

MSE 360: Kinetic Processes in Materials, S2021
Chapter 4: Summary of Transient Diffusion Cases

1) Transient Semi-Infinite Diffusion from Constant Surface Source

B.C.: $C_i(x = 0, t) = C_i^*$

I.C.: $C_i(x > 0, t = 0) = C_i^\circ$

Solution: $C_i(x, t) = C_i^\circ + (C_i^* - C_i^\circ) \operatorname{erfc} \left[\frac{x}{2\sqrt{Dt}} \right]$

2) Transient Interdiffusion in Two Semi-Infinite Bodies

B.C.: $C_i(x = 0, t) = \frac{1}{2} C_A^*$

I.C. 1: $C_A(x > 0, t = 0) = 0$

I.C. 2: $C_A(x < 0, t = 0) = C_A^*$

Solution:

$$C_A(x, t) = \frac{1}{2} C_A^* \operatorname{erfc} \left[\frac{x}{2\sqrt{Dt}} \right]$$

$$C_B(x, t) = \frac{1}{2} C_B^* + \frac{1}{2} C_B^* \operatorname{erf} \left[\frac{x}{2\sqrt{Dt}} \right]$$

3) Transient Infinite Diffusion of a Rectangular Concentration Profile

B.C. 1: $C_i(x = -\infty, t) = 0$

B.C. 2: $C_i(x = \infty, t) = 0$

I.C.: $C_i(-L < x < L, t = 0) = C_i^*$

Solution: $C_i(x, t) = \frac{1}{2} C_i^* \left[\operatorname{erf} \left[\frac{x+L}{2\sqrt{Dt}} \right] - \operatorname{erf} \left[\frac{x-L}{2\sqrt{Dt}} \right] \right]$

4) Transient Infinite Diffusion of a Thin Layer

B.C. 1: $C_i(x = -\infty, t) = 0$

B.C. 2: $C_i(x = \infty, t) = 0$

I.C.: "Amount of Species" i at $x = 0$ and $t = 0$; N_i

Solution: $C_i(x, t) = \frac{N_i}{\sqrt{4\pi Dt}} \exp \left(-\frac{x^2}{4Dt} \right)$

5) **Transient Finite (Symmetric) Planar Diffusion**

B.C. 1: $C_i(x = 0, t) = C_i^*$

B.C. 2: $C_i(x = L, t) = C_i^*$

I.C. : $C_i(0 < x < L, t = 0) = C_i^\circ$

Solution:

$$C_i(x, t) = C_i^* + (C_i^\circ - C_i^*) \frac{4}{\pi} \sum_{j=0}^{\infty} \left(\frac{1}{2j+1} \sin \left[(2j+1) \pi \frac{x}{L} \right] e^{-(2j+1)^2 \pi^2 D t / L^2} \right)$$

6) **Transient Finite (Symmetric) Spherical Diffusion**

B.C. 1: $C_i(r = \pm R, t) = C_i^*$

B.C. 2: $J_i(r = 0, t) = 0$

I.C. : $C_i(-R < r < R, t = 0) = C_i^\circ$

Solution:

$$C_i(x, t) = C_i^\circ + (C_i^* - C_i^\circ) \left[1 + \frac{2R}{\pi r} \sum_{j=1}^{\infty} \left(\frac{(-1)^j}{j} \sin \left[j \pi \frac{r}{R} \right] e^{-(j)^2 \pi^2 D t / R^2} \right) \right]$$